

StemCells Protocol Simulator: a privacy-preserving, in-browser tool for epigenetic-age estimation and transparent regenerative-therapy planning

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Summary

StemCells Protocol Simulator is a browser-based tool that estimates a person's **epigenetic (biological) age** from a DNA-methylation profile and presents it, together with clearly-labelled illustrative therapy-planning models, in a single guided workflow. The epigenetic-age computation is a faithful TypeScript implementation of the Horvath (2013) multi-tissue clock [horvath2013] and runs **entirely on the user's device** — the methylation file is never uploaded to a server. The same code powers a public educational website and an offline research build. This paper describes the tool and validates its epigenetic-age estimates against a public reference cohort (GSE40279, n=656), reproducing reference accuracy: Pearson $r = 0.918$, mean absolute error 4.77 years, median 3.87 years. Only the epigenetic-age computation is validated; the accompanying reprogramming, tumorigenicity and cellular-outcome modules are explicitly illustrative planning heuristics, not validated predictors.

Statement of need

Epigenetic clocks are now central to ageing and regenerative-medicine research, but running them typically requires an R/Python environment and uploading sensitive genomic data to a third-party service. There is a gap for a **zero-install, privacy-preserving** implementation that a clinician, researcher, or informed patient can run in a web browser, and that is explicit about the difference between a *measured* quantity (biological age from a validated clock) and *illustrative* forward-looking models (e.g. partial- reprogramming or cell-therapy projections). StemCells Protocol Simulator fills that gap: it computes epigenetic age client-side from either Illumina array beta values or bisulfite- sequencing (.cov/bedGraph) output, and wraps the result in a transparent, disclaimed planning interface aimed at making users better-informed rather than replacing clinical judgement.

Functionality

- **Input:** Illumina 450K/EPIC array beta CSV (cg IDs) or WGBS/RRBS coverage files

(.cov, bedGraph, bedMethyl); an integrated converter maps genomic-coordinate methylation onto the clock's CpGs (hg38).

- **Epigenetic age:** the Horvath (2013) 353-CpG clock, computed in-browser, with CpG

coverage reporting and age-acceleration relative to chronological age.

- **Illustrative planning models (explicitly not validated):** partial-reprogramming and

tissue-regeneration projections, an OSK/AAV construct or IV-exosome carrier sketch, a per-patient tumorigenicity and immunogenicity risk envelope, and a variant-informed cellular-outcome cartoon. These are deterministic planning heuristics presented as model estimates, **not** measured or clinical outcomes.

- **Output:** an on-screen animated run and an exportable PDF report.

- **Privacy:** all computation is client-side; no methylation data leaves the browser.

Validation

We validated the epigenetic-age estimator — the tool's measured core — against **GSE40279** [@hannum2013], a public whole-blood Illumina 450K cohort of 656 individuals with known chronological ages, using the bundled scripts/ bench - clock . ts harness, which calls the **same** production `predict()` function the application uses.

Metric	Value
Samples evaluated	656
Mean CpG coverage	100%
Pearson r (predicted vs chronological age)	0.918
Mean absolute error	4.77 years
Median absolute error	3.87 years

These figures are consistent with the accuracy Horvath (2013) reports for this clock on healthy cohorts (median error ~ 3.6 years), confirming that the in-browser implementation reproduces the reference method. A predicted-vs-chronological-age scatter plot is shown in \autoref{fig:validation}.

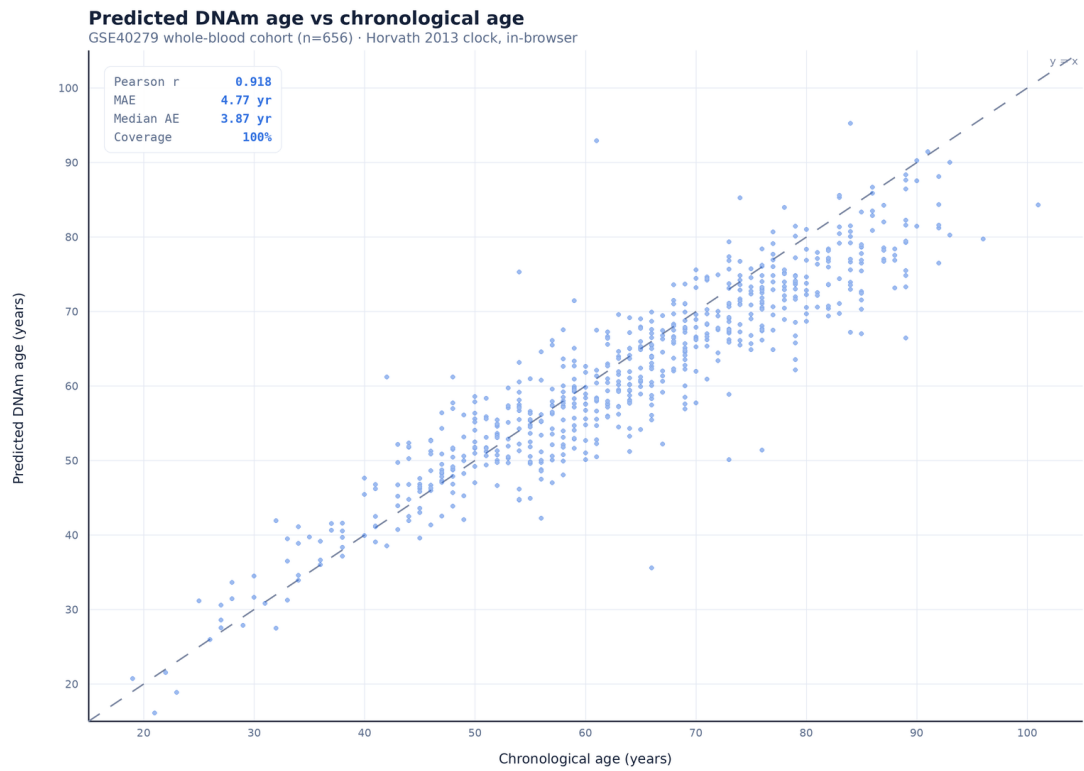


Figure 1. Predicted DNAm age vs chronological age on GSE40279 (n=656). Dashed line is $y=x$.

Limitations

Only the **epigenetic-age computation** is validated here. The reprogramming, tumorigenicity, immunogenicity, and cellular-outcome modules are illustrative planning heuristics, not validated predictors, and the tool labels them as such throughout; they must not be interpreted as clinical results or treatment recommendations. The current release implements a single clock (Horvath 2013); multi-clock support (e.g. Hannum, PhenoAge) is planned to remove single-estimator dependence.

Availability

Source code and the validation harness are available at <<https://github.com/sanjaydoc/Stemcellsprotocol>>; the tool runs at <<https://stemcellsprotocol.com>>. This release is archived on Zenodo (DOI: 10.5281/zenodo.22760374 (<https://doi.org/10.5281/zenodo.22760374>)). The validation dataset GSE40279 is available from NCBI GEO.

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References

References

- Horvath, Steve (2013). DNA methylation age of human tissues and cell types. *Genome Biology*. doi:10.1186/gb-2013-14-10-r115
- Hannum, Gregory et al. (2013). Genome-wide methylation profiles reveal quantitative views of human aging rates. *Molecular Cell*. doi:10.1016/j.molcel.2012.10.016
- Levine, Morgan E. et al. (2018). An epigenetic biomarker of aging for lifespan and healthspan (PhenoAge). *Aging*. doi:10.18632/aging.101414
- Lu, Yuancheng et al. (2020). Reprogramming to recover youthful epigenetic information and restore vision. *Nature*. doi:10.1038/s41586-020-2975-4